

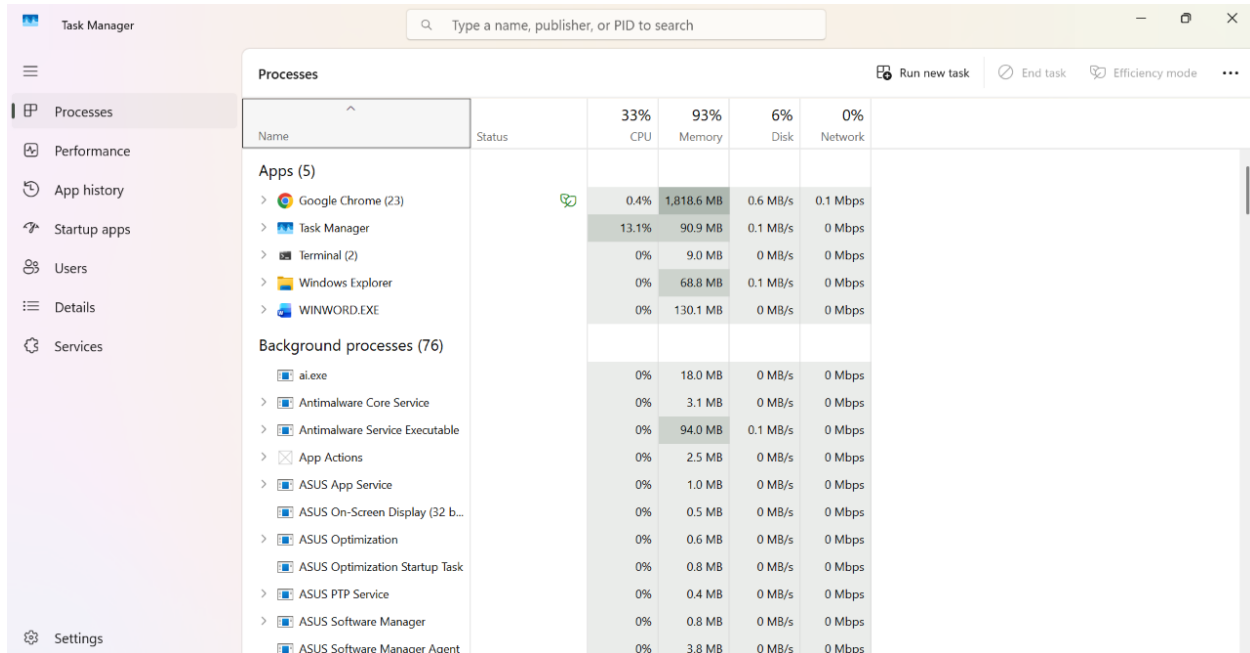
Basic System

Monitoring Using Windows Task Manager

Objective: To observe the current performance of a computer system by checking CPU, memory, disk, and network usage.

Procedure:

1. Open **Task Manager** by pressing **Ctrl + Shift + Esc**.

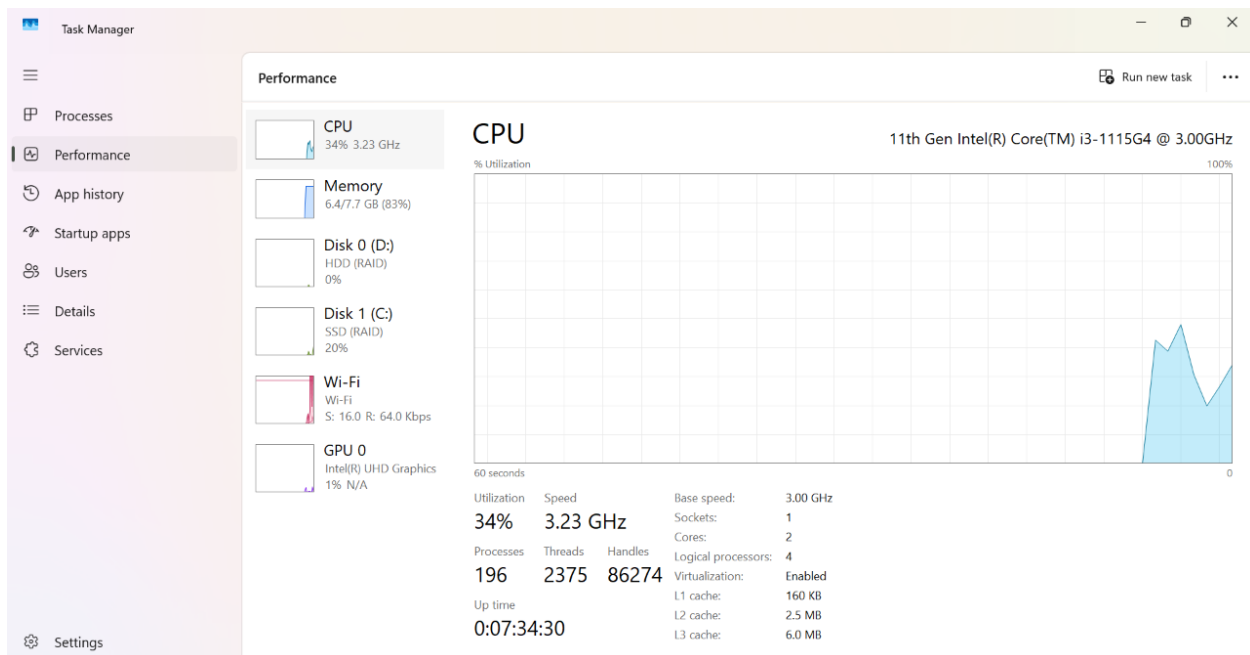


Processes		33% CPU	93% Memory	6% Disk	0% Network
Name	Status				
Apps (5)					
> Google Chrome (23)		0.4%	1,818.6 MB	0.6 MB/s	0.1 Mbps
> Task Manager		13.1%	90.9 MB	0.1 MB/s	0 Mbps
> Terminal (2)		0%	9.0 MB	0 MB/s	0 Mbps
> Windows Explorer		0%	68.8 MB	0.1 MB/s	0 Mbps
> WINWORD.EXE		0%	130.1 MB	0 MB/s	0 Mbps
Background processes (76)					
ai.exe		0%	18.0 MB	0 MB/s	0 Mbps
> Antimalware Core Service		0%	3.1 MB	0 MB/s	0 Mbps
> Antimalware Service Executable		0%	94.0 MB	0.1 MB/s	0 Mbps
> App Actions		0%	2.5 MB	0 MB/s	0 Mbps
> ASUS App Service		0%	1.0 MB	0 MB/s	0 Mbps
> ASUS On-Screen Display (32 b...		0%	0.5 MB	0 MB/s	0 Mbps
> ASUS Optimization		0%	0.6 MB	0 MB/s	0 Mbps
> ASUS Optimization Startup Task		0%	0.8 MB	0 MB/s	0 Mbps
> ASUS PTP Service		0%	0.4 MB	0 MB/s	0 Mbps
> ASUS Software Manager		0%	0.8 MB	0 MB/s	0 Mbps
> ASUS Software Manager Agent		0%	3.8 MB	0 MB/s	0 Mbps

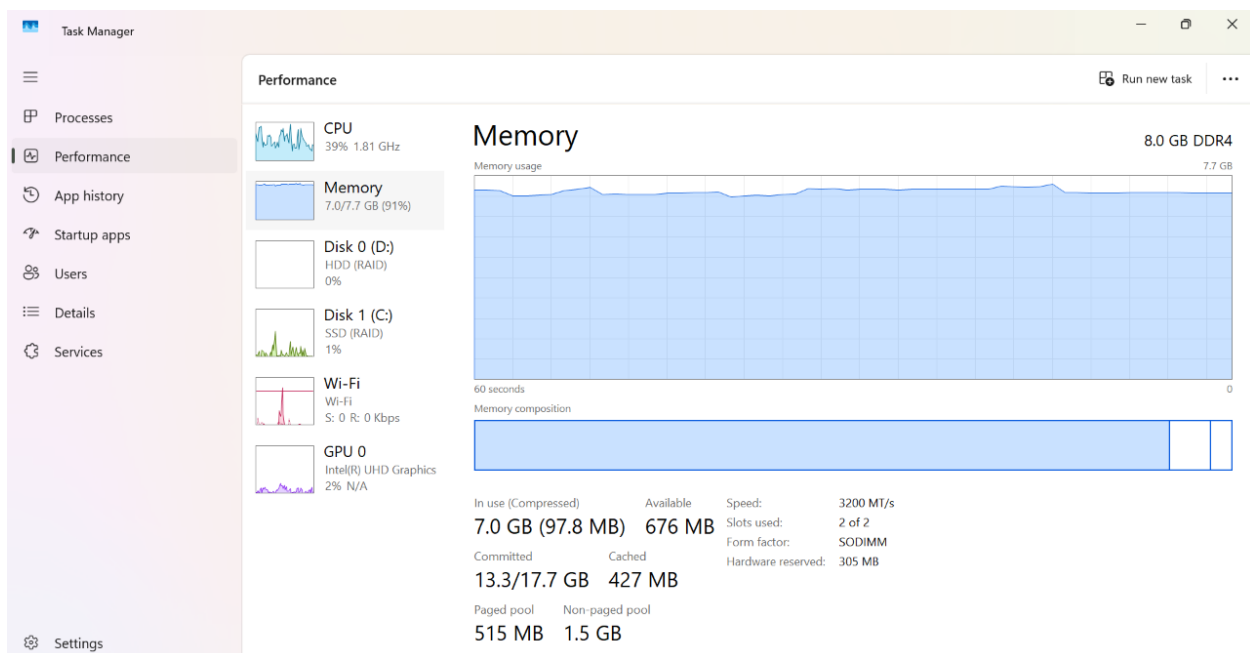
2. Select the Performance tab.

3. Observe the following:

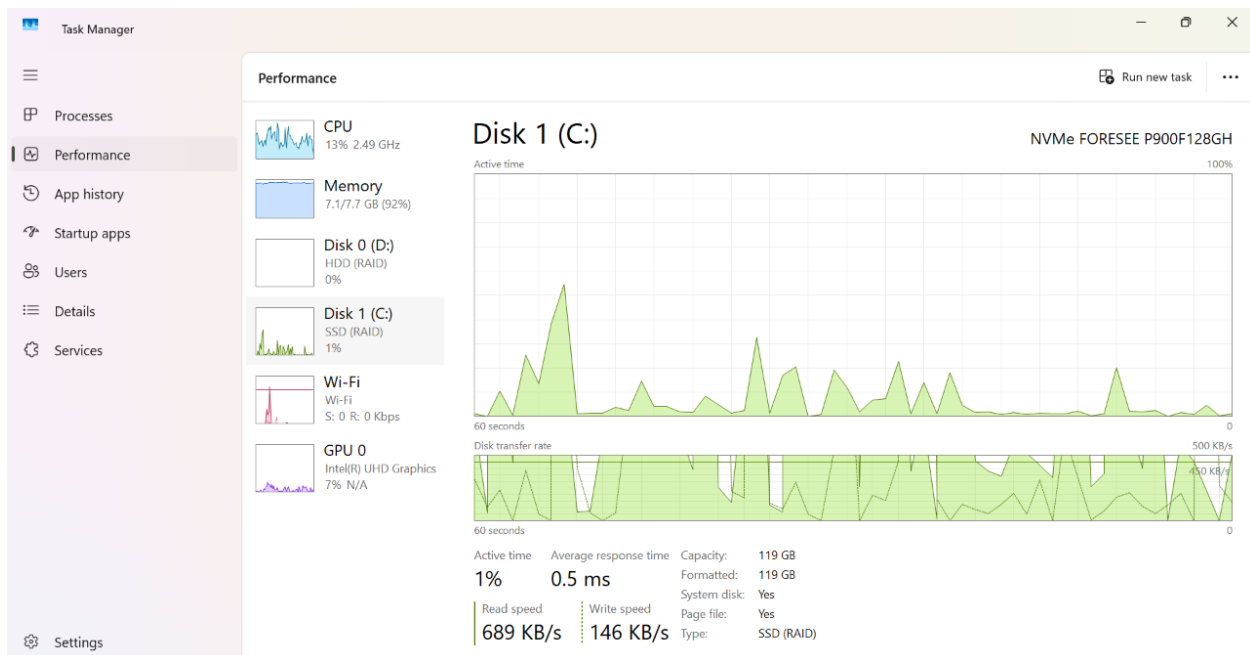
a. **CPU:** Shows how much processing power is being used.



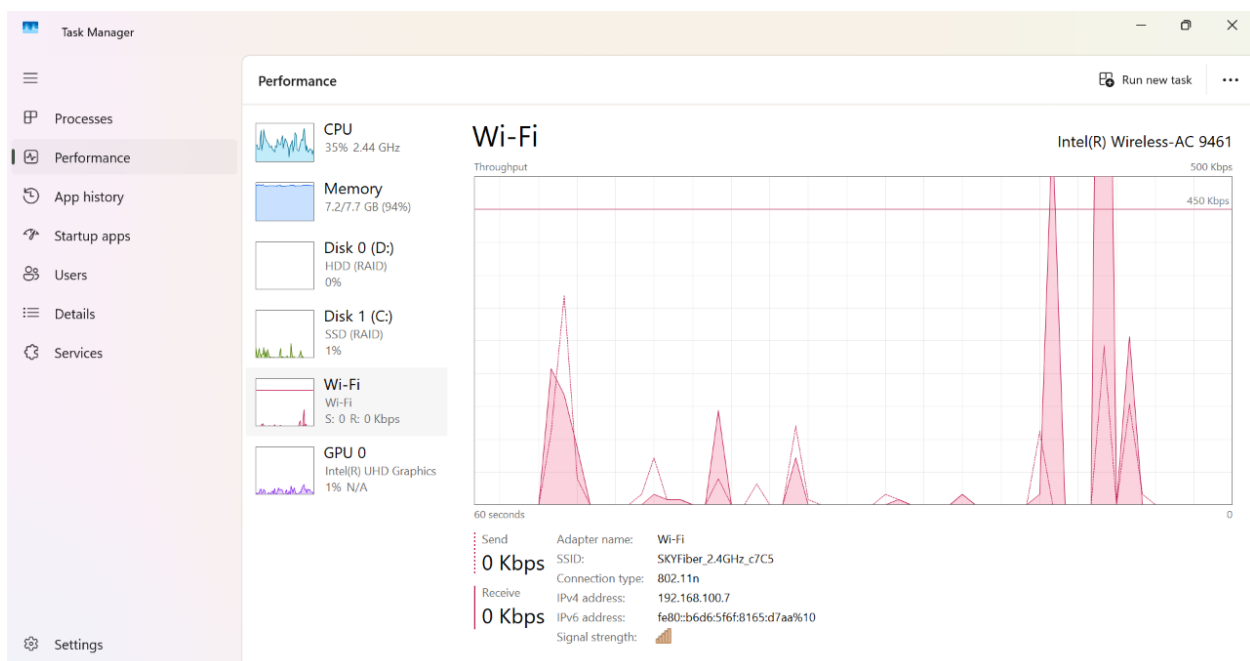
b. **Memory:** Shows how much RAM is currently occupied.



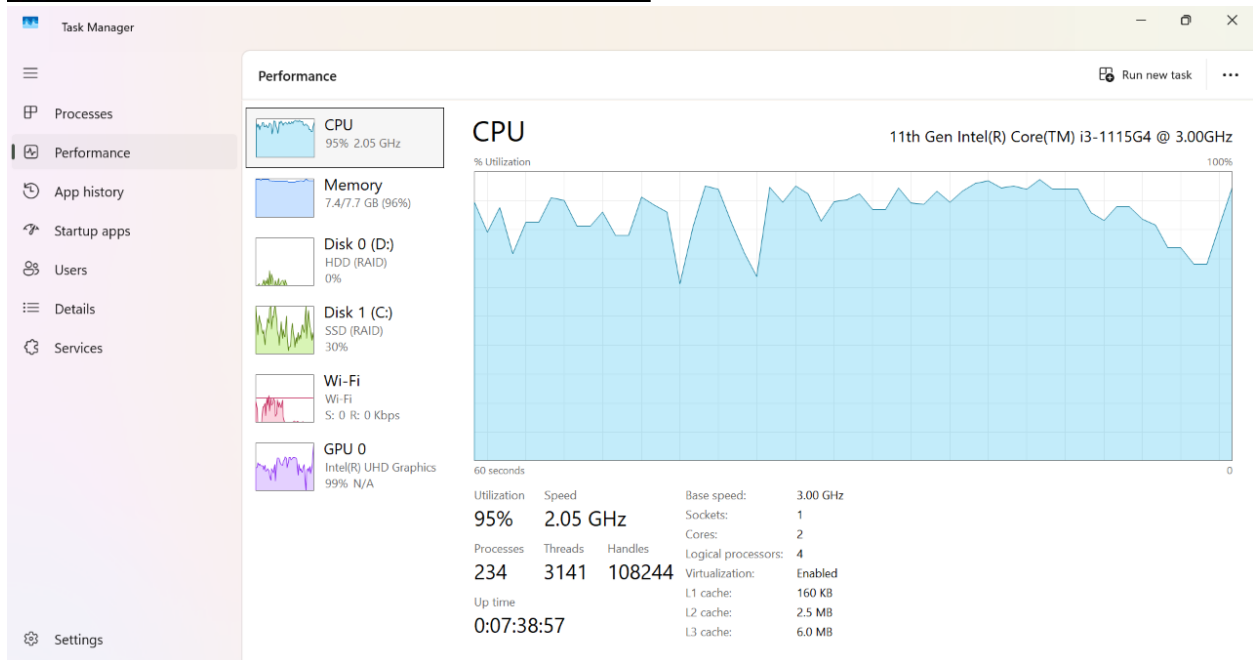
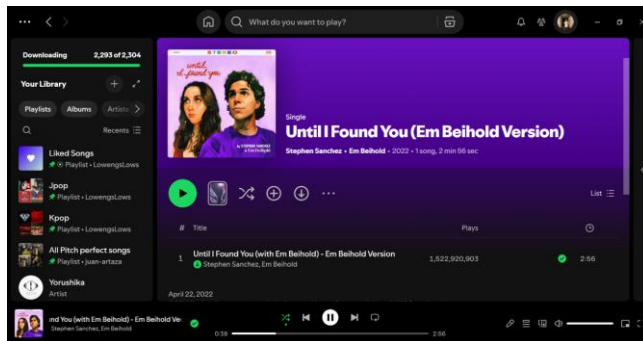
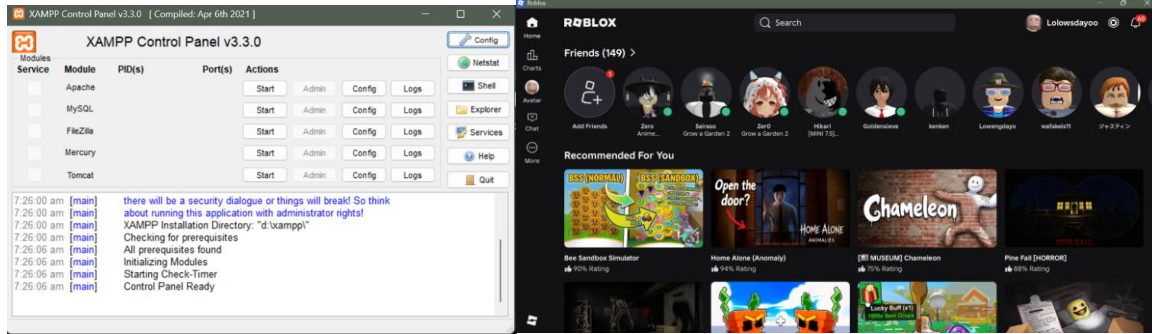
c. **Disk:** Shows storage activity.



4. Network: Shows data transfer activity.



5. Open several applications and observe how the values change.



6. Record which resources increase when more applications are running.

Name	Status	93% CPU	93% Memory	7% Disk	12% Network
Apps (7)					
> Google Chrome (15)		0.4%	851.2 MB	0.1 MB/s	0 Mbps
> Microsoft Word		0.4%	79.3 MB	0.1 MB/s	0 Mbps
> Roblox Game Client		49.1%	991.3 MB	8.2 MB/s	0.1 Mbps
> Spotify (7)		2.6%	446.6 MB	0.1 MB/s	0.1 Mbps
> Task Manager		6.3%	56.1 MB	0.1 MB/s	0 Mbps
> Windows Explorer		3.3%	58.6 MB	0.1 MB/s	0 Mbps
> xampp-control.exe (32 bit)		1.1%	1.5 MB	0 MB/s	0 Mbps
Background processes (72)					
> Antimalware Core Service		0%	2.2 MB	0 MB/s	0 Mbps
> Antimalware Service Executable		0.4%	156.8 MB	0.2 MB/s	0 Mbps
> ASUS App Service		0%	1.5 MB	0 MB/s	0 Mbps
> ASUS On-Screen Display (32 b...		0%	1.4 MB	0 MB/s	0 Mbps
> ASUS Optimization		0%	1.0 MB	0 MB/s	0 Mbps
> ASUS Optimization Startup Task		0%	0.8 MB	0 MB/s	0 Mbps
> ASUS PTP Service		0%	0.4 MB	0 MB/s	0 Mbps
> ASUS Software Manager		0%	0.8 MB	0 MB/s	0 Mbps
> ASUS Software Manager Agent		0%	4.9 MB	0 MB/s	0 Mbps

Roblox Application Uses more of the Cpu than any other application.

Expected Result: Students should observe that computer resources change depending on the tasks being performed. High resource usage may indicate the need for optimization or further investigation (Microsoft, 2024).

Checking Hardware and Network Performance

Objective: This activity demonstrates how IT personnel observe system health before problems affect users and operations (Microsoft, 2024).

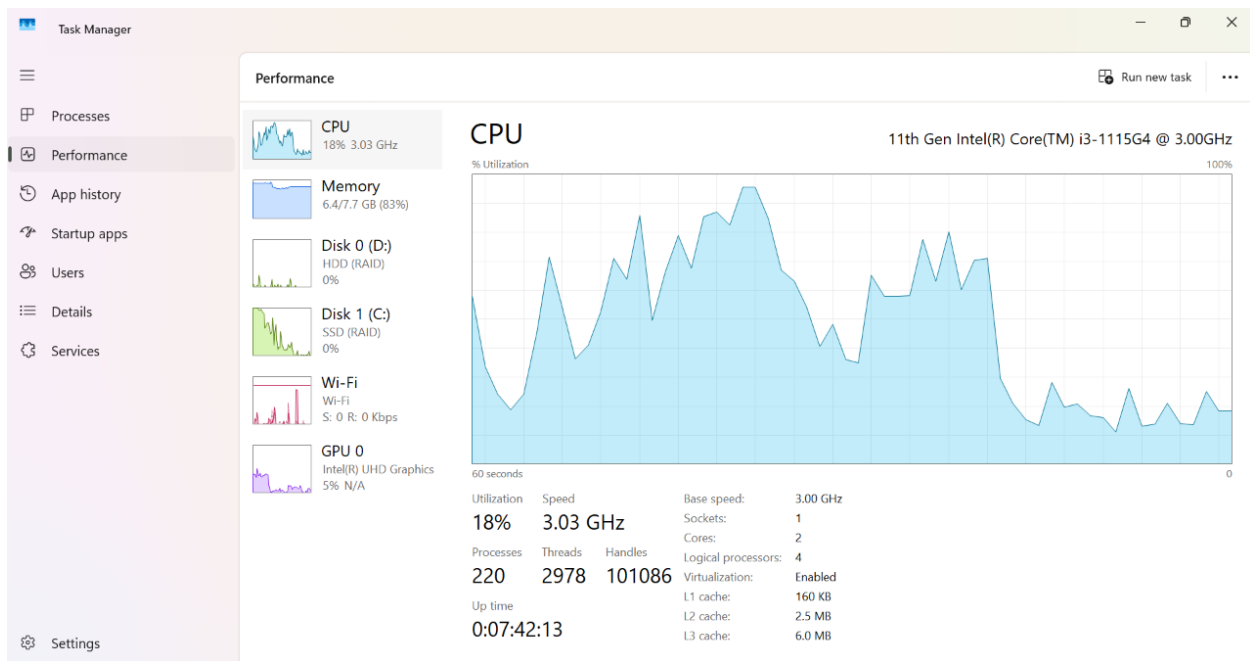
Tools:

- Windows Task Manager
- Windows Performance Monitor

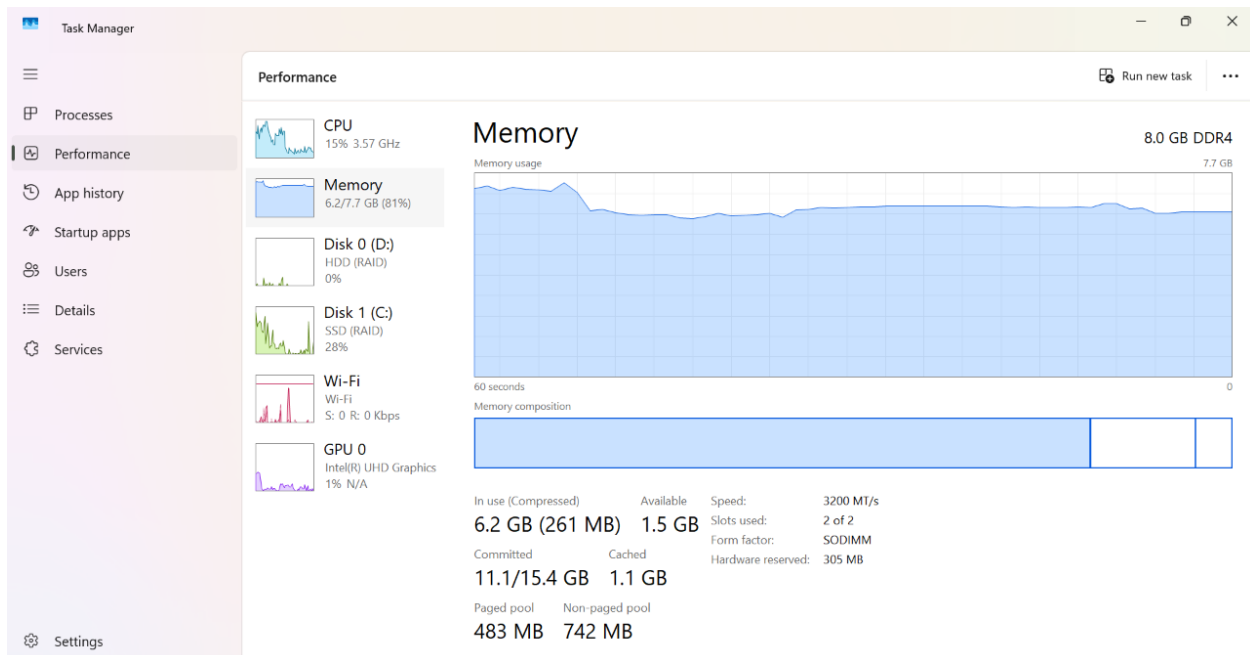
Procedure:

1. Open Task Manager.
2. Select the Performance tab.
3. Check:

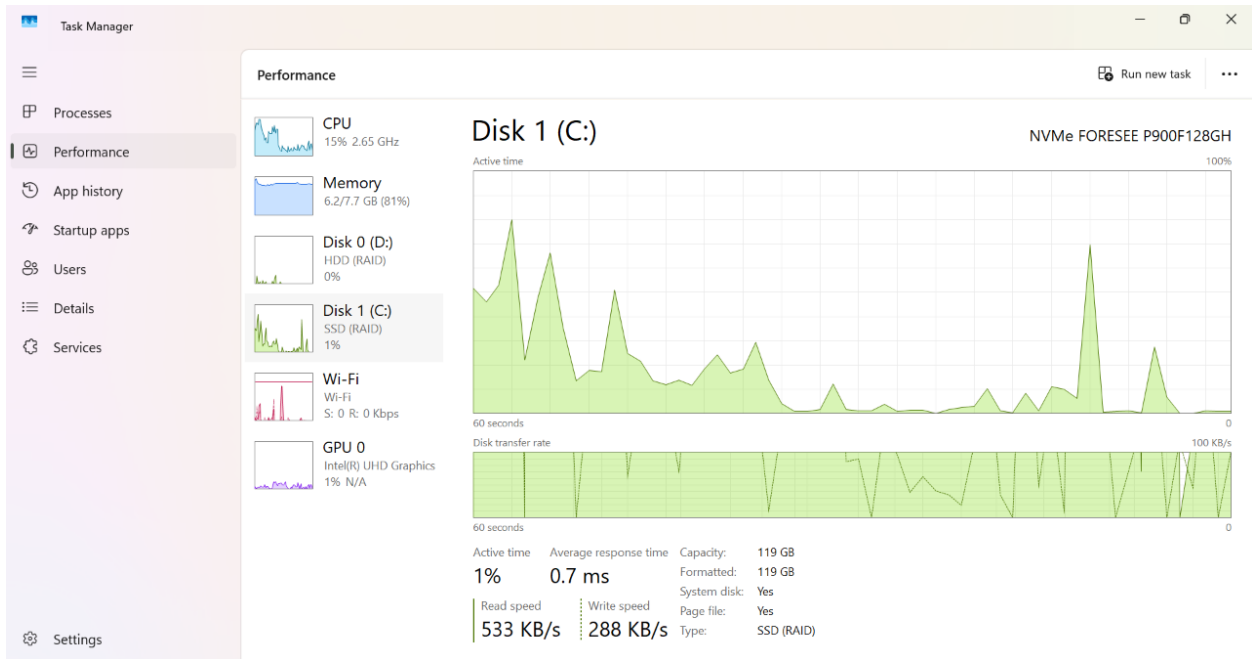
a. CPU utilization



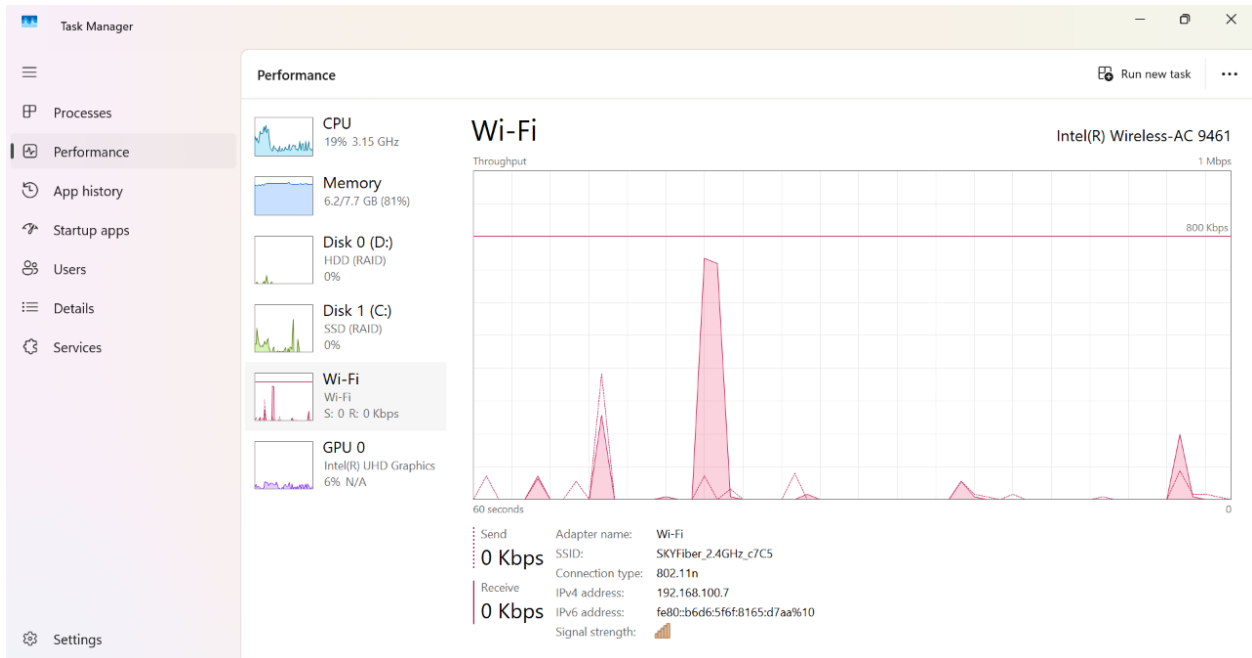
b. Memory usage



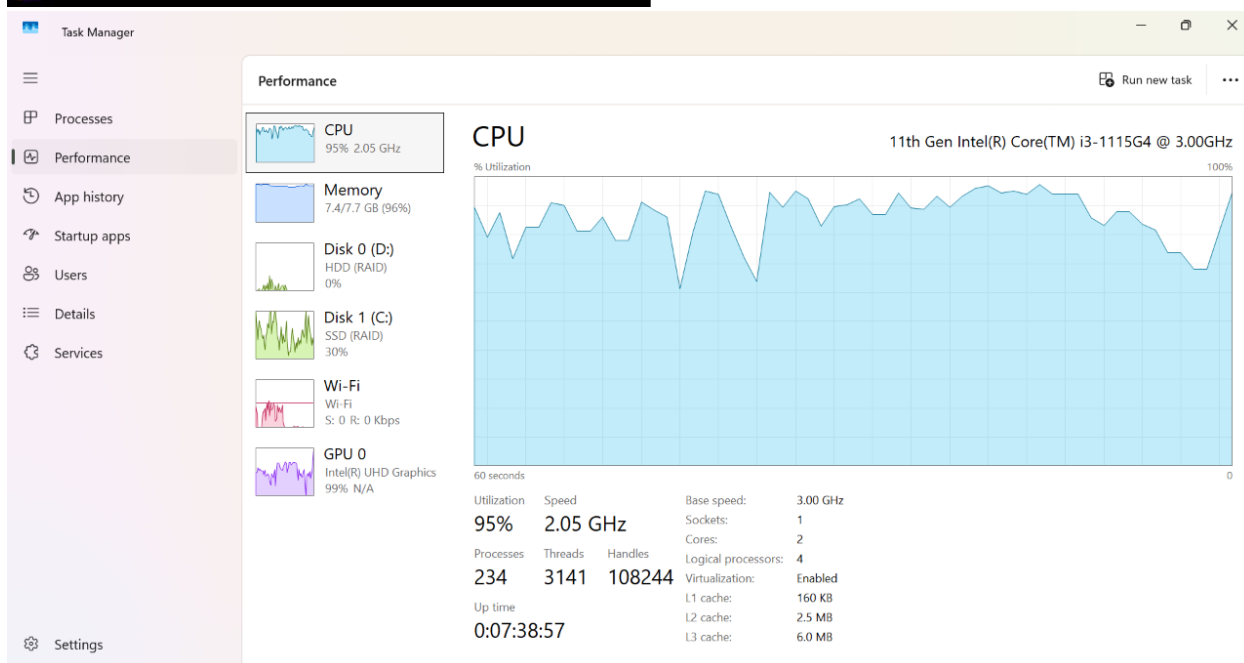
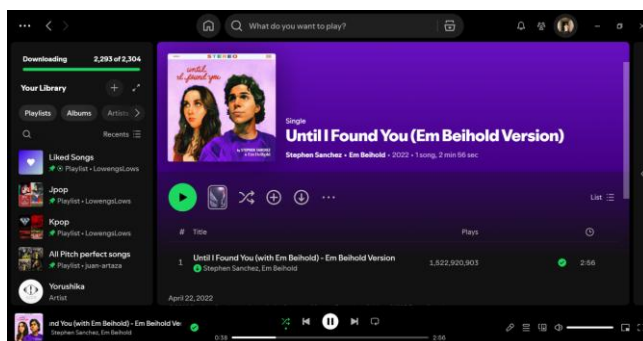
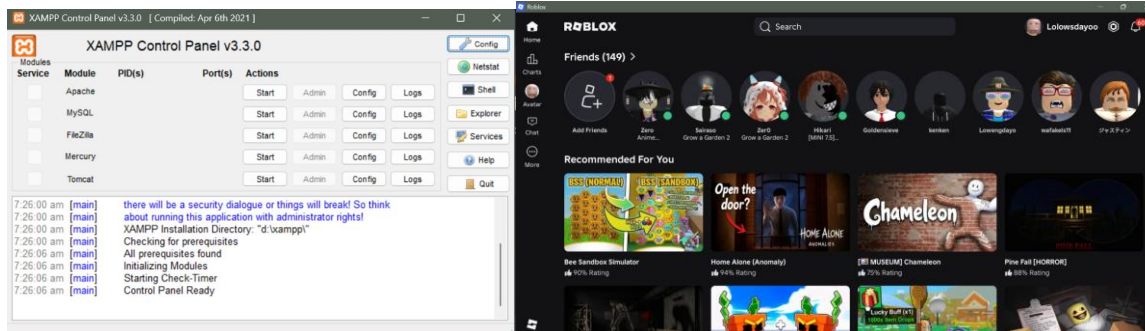
c. Disk activity



d. Network activity



4. Open several applications and observe changes.



5. Record the highest resource usage observed.

The screenshot shows the Windows Task Manager interface. The 'Processes' tab is selected, displaying a list of running applications and background services. The top of the window shows a search bar and buttons for 'Run new task', 'End task', and 'Efficiency mode'. The main table lists processes with columns for Name, Status, CPU, Memory, Disk, and Network usage.

Name	Status	93% CPU	93% Memory	7% Disk	12% Network
Apps (7)					
Google Chrome (15)	Running	0.4%	851.2 MB	0.1 MB/s	0 Mbps
Microsoft Word	Running	0.4%	79.3 MB	0.1 MB/s	0 Mbps
Roblox Game Client	Running	49.1%	991.3 MB	8.2 MB/s	0.1 Mbps
Spotify (7)	Running	2.6%	446.6 MB	0.1 MB/s	0.1 Mbps
Task Manager	Running	6.3%	56.1 MB	0.1 MB/s	0 Mbps
Windows Explorer	Running	3.3%	58.6 MB	0.1 MB/s	0 Mbps
xampp-control.exe (32 bit)	Running	1.1%	1.5 MB	0 MB/s	0 Mbps
Background processes (72)					
Antimalware Core Service	Running	0%	2.2 MB	0 MB/s	0 Mbps
Antimalware Service Executable	Running	0.4%	156.8 MB	0.2 MB/s	0 Mbps
ASUS App Service	Running	0%	1.5 MB	0 MB/s	0 Mbps
ASUS On-Screen Display (32 b...	Running	0%	1.4 MB	0 MB/s	0 Mbps
ASUS Optimization	Running	0%	1.0 MB	0 MB/s	0 Mbps
ASUS Optimization Startup Task	Running	0%	0.8 MB	0 MB/s	0 Mbps
ASUS PTP Service	Running	0%	0.4 MB	0 MB/s	0 Mbps
ASUS Software Manager	Running	0%	0.8 MB	0 MB/s	0 Mbps
ASUS Software Manager Agent	Running	0%	4.9 MB	0 MB/s	0 Mbps

Guide Questions:

1. Which resource increased the most when applications were opened?

Based on the Task Manager screenshot, the CPU and Memory (RAM) usage increased the most, with both reaching **93%**. This shows that the opened applications, particularly Roblox Game Client, along with Google Chrome, Spotify, and Microsoft Word, were using a large amount of the computer's processing power and memory. In comparison, Disk usage remained at 7% and Network usage at 12%, indicating that they were not under heavy load.

2. What signs may indicate that a computer is experiencing performance problems?

A computer may be experiencing performance problems when it becomes slow to respond, applications take longer to open, or programs freeze and stop responding. High CPU and memory usage, like the 93% shown in the Task Manager, can also cause lag, delayed loading times, and reduced system performance. If these high usage levels continue for an extended period, the computer may struggle to run multiple applications efficiently.

3. Why is monitoring important before a system failure occurs?

Monitoring system resources is important because it helps identify performance issues before they lead to a system failure. By regularly checking CPU, memory, disk, and network usage, users can determine which applications are consuming the most resources and take action, such as closing unnecessary programs or upgrading hardware if needed. This helps prevent crashes, freezes, and possible data loss while keeping the computer running smoothly and efficiently.

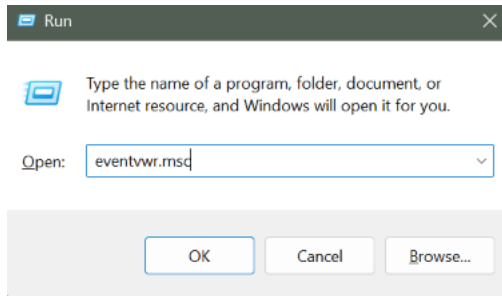
Viewing System Logs Using Windows Event Viewer

Objective: To identify and interpret basic system events recorded by a computer.

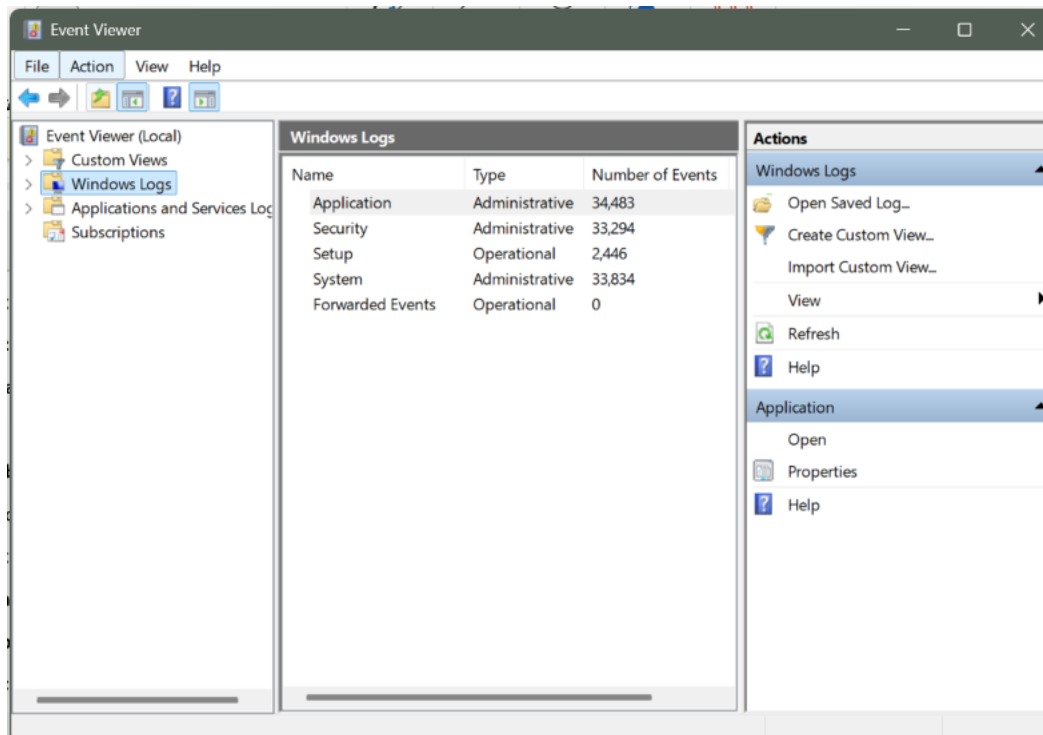
Tool: Windows Event Viewer

Procedure:

1. Press Windows Key + R.
2. Type eventvwr.msc and press Enter.

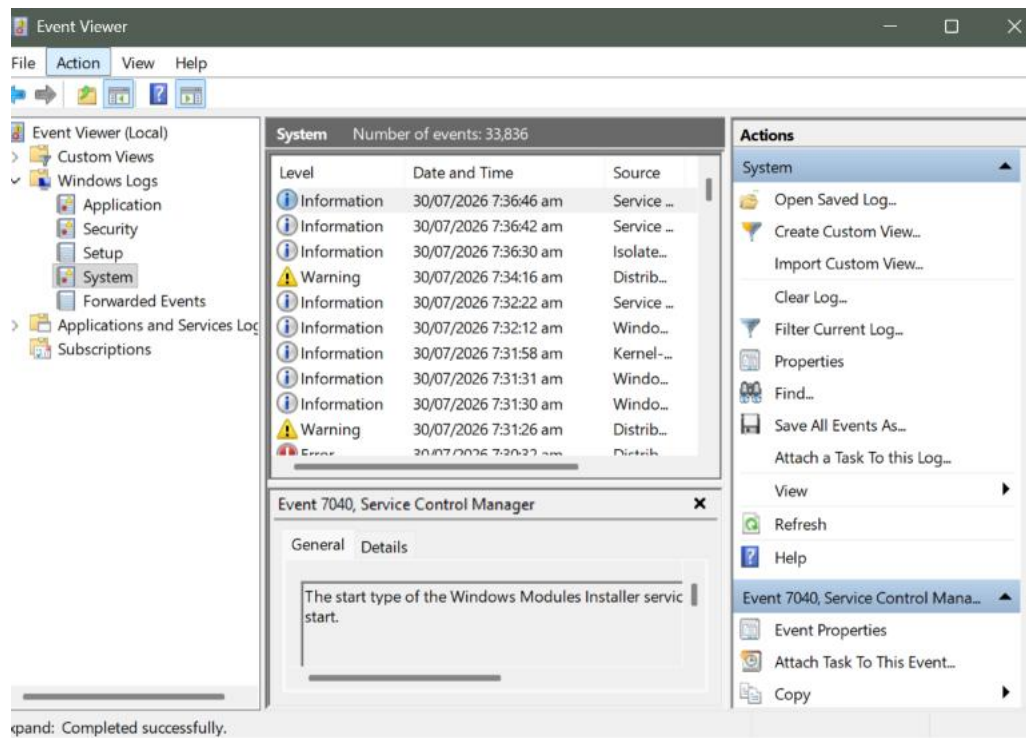


3. Select Windows Logs.



4. Open:

a. System Logs – for operating system activities



The screenshot shows the Windows Event Viewer application. The left-hand pane displays the tree view with 'Event Viewer (Local)' expanded, showing 'Custom Views', 'Windows Logs', and 'Applications and Services Logs'. Under 'Windows Logs', 'Security' is selected. The main pane, titled 'Security', shows a list of 11 events. All events have the source 'Microsoft Windows security auditing' and ID 537. The details pane for event 5379 is open, showing the message 'Credential Manager credentials were read.' and the subject 'Security ID: LAPTOP-VQMUA'.

Keyword	Date and Time	Source	Event ID
Audit...	30/07/2026 7:37:45 am	Micros...	537
Audit...	30/07/2026 7:37:45 am	Micros...	537
Audit...	30/07/2026 7:37:45 am	Micros...	537
Audit...	30/07/2026 7:37:45 am	Micros...	537
Audit...	30/07/2026 7:37:45 am	Micros...	537
Audit...	30/07/2026 7:37:45 am	Micros...	537
Audit...	30/07/2026 7:37:45 am	Micros...	537
Audit...	30/07/2026 7:37:45 am	Micros...	537
Audit...	30/07/2026 7:37:40 am	Micros...	537
Audit...	30/07/2026 7:37:40 am	Micros...	537
Audit...	30/07/2026 7:37:40 am	Micros...	537

Event 5379, Microsoft Windows security auditing.

General Details

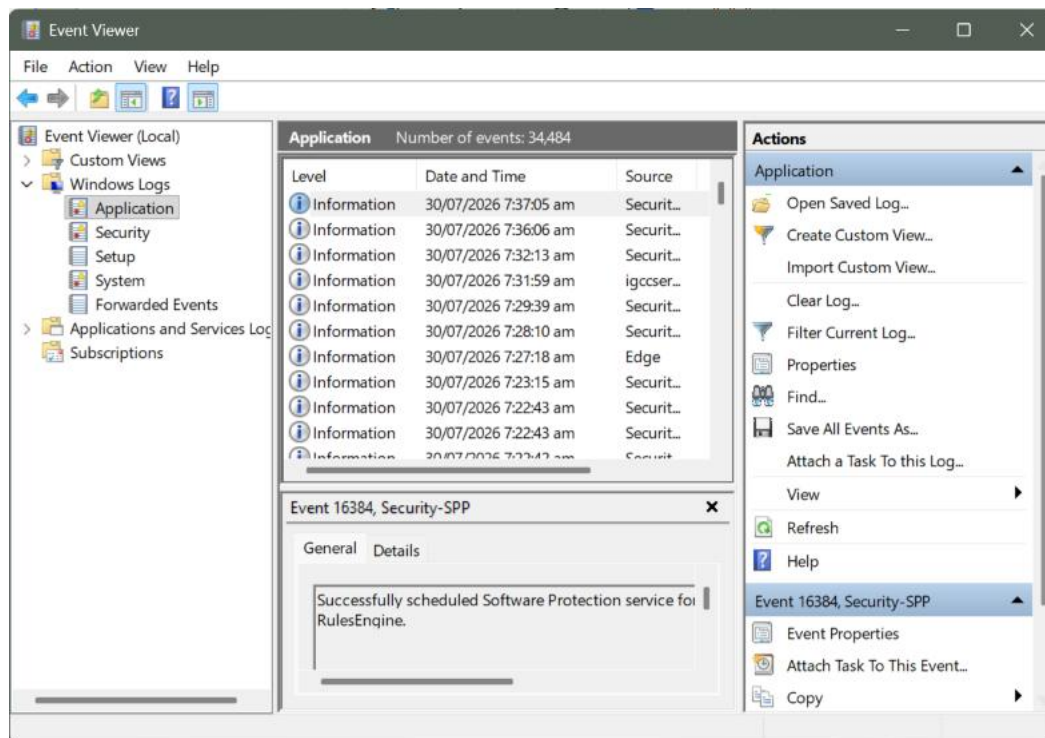
Credential Manager credentials were read.

Subject: Security ID: LAPTOP-VQMUA

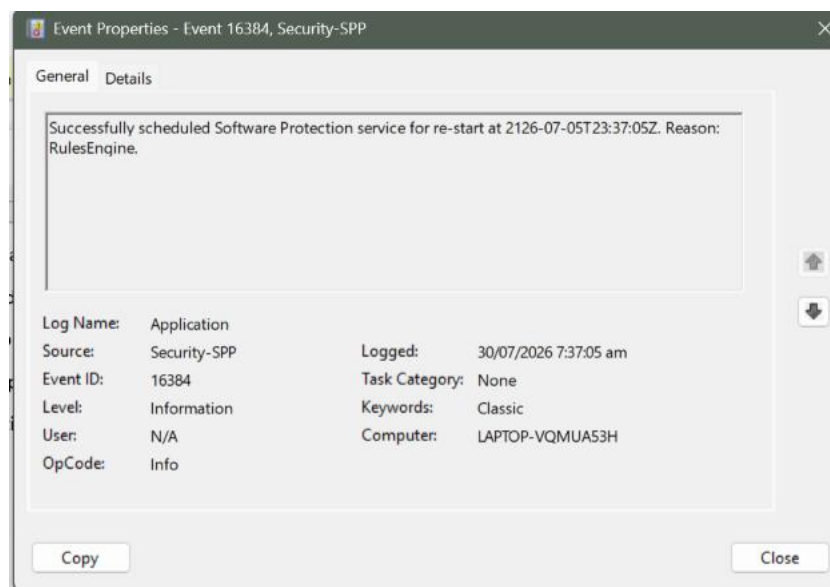
Actions

- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Clear Log...
- Filter Current Log...
- Properties
- Find...
- Save All Events As...
- Attach a Task to This Log...
- View
- Refresh
- Help
- Event 5379, Microsoft Windows security auditing...
- Event Properties
- Attach Task To This Event...
- Copy

c. Application Logs – for software-related events



5. Select an event and observe:



- a. **Date and time**
30/07/2026 7:37:05 am
- b. **Event source**
Security-SPP
- c. **Event type**
Information level and Application log name

d. Description of the activity

Successfully scheduled Software Protection service for re-start at 2126-07-05T23:37:05Z. Reason: RulesEngine.